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Human-Computer Interaction 2

Evaluating and Redesigning the MyMapúa Portal for Mobile Usability

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PROJECT DESCRIPTION

Project Overview

The MyMapúa Portal Mobile Usability Evaluation and Redesign project focuses on assessing and enhancing the mobile user experience of Mapúa University's portal. Originally designed for desktop screens, the current portal presents usability bottlenecks, such as cluttered layouts, complex navigation, and unoptimized touch targets when accessed via mobile devices. This project redesigned the Mapúa University portal with a mobile-first approach, developed a prototype, and evaluated the redesigned interface through usability testing and surveys with 15 Mapúa students to assess its effectiveness and ease of use for accessing grades, schedules, and curriculum progress.

Background of the Study

Higher education institutions rely on student portals to give access to essential academic and administrative services, such as grade tracking, schedule viewing, curriculum checklists, and statements of account (Purdue University, 2023; University of Arizona, n.d.). However, many of these systems, including the MyMapúa student portal, were originally built for desktop interfaces. They were later brought to mobile screens without being fully adapted for mobile constraints.

When using mobile devices, students run into usability problems caused by limited screen real estate, crowded visual layouts, hidden submenus, and complex navigation paths. These interface flaws cause cognitive overload, increase task completion time, and make routine tasks harder to finish on mobile phones (Faudzi et al., 2023). While student portals are widely used, research specifically on their mobile usability remains limited. Evaluating the MyMapúa mobile interface helps address this gap by identifying key usability bottlenecks and guiding the development of a user-centered mobile prototype.

Purpose of the Study

This study aims to assess and improve the accessibility of essential academic services on mobile devices. By identifying key interface bottlenecks and creating a mobile-first prototype, the project seeks to establish a seamless user experience for students. Testing the prototype with 15 Mapúa students allows the project to validate design enhancements and demonstrate how optimized navigation and layout structures improve efficiency for routine student tasks.

Problem Statement

The MyMapúa Portal Mobile Usability Evaluation and Redesign project addresses the difficulties students encounter when using the MyMapúa portal on mobile devices. The existing interface was primarily designed for desktop use, which can make important features harder to access and navigate on smaller screens. To address these concerns, the project applies a mobile-first design approach to create a redesigned prototype that simplifies

navigation, improves layout organization, and optimizes touch interactions. The redesigned prototype is then tested and evaluated by 15 Mapúa students to determine how well it supports common academic tasks, including viewing grades, schedules, profile and curriculum progress.

Research Objectives

General Objective

To redesign and evaluate the MyMapúa portal using a mobile-first approach to improve the usability and user experience of accessing academic and administrative information on mobile devices.

Specific Objectives

- To analyze user friction and interface bottlenecks experienced by Mapúa students on the existing mobile portal.
- To redesign the portal interface using a mobile-first approach based on the identified usability issues and user needs.
- To develop a mobile prototype that provides improved navigation, layout, and touch interactions for common portal tasks.
- To evaluate the redesigned prototype through usability testing with 15 Mapúa students.
- To assess the students' perceived ease of use, effectiveness, and overall satisfaction with the redesigned mobile interface.

Primary Users

The primary users of the redesigned MyMapúa mobile portal are **Mapúa University students** who access the portal through mobile devices to manage and view their academic and administrative information. This includes students who regularly use the portal to check their grades, view class schedules, monitor curriculum progress, and manage their accounts.

Scope of the Study

The study focuses on the mobile usability, interface design, and user experience of the MyMapúa student portal. It covers the redesign of selected portal features using a mobile-first approach, including grades, class schedules, curriculum progress, and profile. A mobile prototype was developed based on identified usability concerns and evaluated through usability testing and surveys involving 15 Mapúa University students. The evaluation focuses on aspects such as navigation, layout, touch interactions, ease of use, effectiveness, and overall user experience.

The study is limited to the prototype and its usability evaluation and does not involve the development or modification of the actual MyMapúa system. It also does not cover the portal's backend systems, database, security infrastructure, or implementation for production use.

Expected Output

The study is expected to produce a mobile-first prototype of the redesigned MyMapúa portal that provides a more organized, accessible, and user-friendly interface for Mapúa students. The prototype will include improved navigation and layouts for accessing grades, class schedules, curriculum progress, and profile. The usability testing and survey results are expected to provide insights into the prototype's ease of use, effectiveness, and overall user experience.

USABILITY TEST PLAN

Version 1.0

Document Overview

This document describes a test plan for conducting a usability test during the development of the redesigned MyMapúa prototype. The goals of usability testing include establishing a baseline of user performance, establishing and validating user performance measures, and identifying potential design concerns to be addressed in order to improve the efficiency, productivity, and end-user satisfaction by resolving mobile viewport constraints, nested navigation menus, and data density issues.

The usability test objectives are:

- To determine design inconsistencies and usability problem areas within the user interface and content areas. Potential sources of error may include:
 - Navigation errors – failure to locate functions, excessive keystrokes to complete a function, failure to follow recommended screen flow.
 - Presentation errors – failure to locate and properly act upon desired information in screens, selection errors due to labeling ambiguities.
 - Control usage problems – improper toolbar or entry field usage.
- Exercise the application or web site under controlled test conditions with representative users. Data will be used to assess whether usability goals regarding an effective, efficient, and well-received user interface have been achieved.
- Establish baseline user performance and user-satisfaction levels of the user interface for future usability evaluations.

The redesigned MyMapúa prototype deploys to active Mapúa University students who access student information services on mobile phones. The user group participating in this usability test consists of active Mapúa students. A minimum of fifteen participants will complete the scenario testing and usability scoring rubric. Testing will occur via online on a forms link for respondents to do it in their own time, which will be conducted between September 28, 2026, and October 1, 2026.

Executive Summary

This usability test evaluates the mobile-first prototype of the MyMapúa student portal designed to replace the legacy desktop-adapted interface. Testing evaluates five core student functions: (1) authenticating and viewing enrolled course schedules on the mobile dashboard, (2) inspecting itemized course grades and score breakdowns, (3) retrieving cumulative GPA or GWA standing, (4) verifying graduation units via the curriculum tracker, and (5) viewing billing charges and payment options on the Statement of Account. The usability goals set a 100% completion rate, an 80% error-free rate, an average time-on-task under 45 seconds, and a System Usability Scale score above 70.

Upon review of this usability test plan, including the draft task scenarios and usability goals for the redesigned MyMapúa prototype, documented acceptance of the plan is expected.

Methodology

The usability test will evaluate fifteen active Mapúa University student participants in individual test sessions lasting approximately 30 minutes each. Sessions take place through remote testing environments using standard mobile web browsers on smartphones. Measures collected include pre-test demographic profiles, task completion success rates, error counts, scenario completion duration (time on task), System Usability Scale scores, the standardized 45-item usability rubric across 10 interface categories, and qualitative suggestions for interface improvement.

Participants

A total of 15 test participants will be recruited through purposive sampling across undergraduate academic programs at Mapúa University. This size follows Nielsen (2000), who found that about 15 users are enough to uncover nearly all usability problems in a single test. Eligibility requires participants to be active students who regularly access the current MyMapúa portal on a mobile phone at least once per academic term. Participants possess basic digital literacy with mobile web applications and touch navigation.

The participants' responsibilities will be to attempt to complete a set of representative task scenarios presented to them in as efficient and timely a manner as possible, and to provide feedback regarding the usability and acceptability of the user interface. The participants will be directed to provide honest opinions regarding the usability of the application, and to participate in post-session subjective questionnaires and debriefing.

The evaluation team will select test participants who meet the stated academic enrollment criteria through direct peer contact within the School of Information Technology. Participants have prior familiarity with legacy MyMapúa features such as checking grades and paying tuition, but have zero prior exposure to the redesigned mobile prototype or screen flows.

Training

Participants will receive a MS Form link that explains what the web app is all about. The participants will receive an overview of the usability test procedure, equipment and software. Participants will know that backend payment gateway processing and official transcript generation operate as prototype simulations without altering official university databases.

Procedure

Remote Testing

Participants will take part in the usability test remotely through an unmoderated online questionnaire accessed through a Microsoft Forms link on their personal mobile devices. Participants complete the evaluation independently at their own convenient time in their everyday setting.

The introductory section of the form briefs participants and instructs them that they are evaluating the web application, rather than the team evaluating the participant. Participants complete a pretest demographic questionnaire and confirm digital informed consent. Consent acknowledges that participation is voluntary, that participation can cease at any time, and that submitted responses remain confidential.

Participants review each task prompt sequentially, open the deployed mobile prototype URL, execute the specified task, and return to the form to log completion status and perceived difficulty. The form instructs participants to complete each task directly before exploring secondary screens. After finishing all five tasks, participants complete the System Usability Scale instrument and provide suggestions regarding layout, navigation, and visual clarity.

Roles

The roles involved in a usability test are as follows. An individual may play multiple roles and tests may not require all roles.

Trainer

- Provide training overview prior to usability testing through standardized written instructions on the online form.

Facilitator

- Provides overview of study to participants
- Defines usability and purpose of usability testing to participants
- Distributes test links and responds to participant technical questions.

Data Logger

- Records participant's actions and comments

Test Observers

- Silent observer
- Assists the data logger in identifying problems, concerns, coding bugs, and procedural errors.
- Reviews qualitative survey submissions.

Test Participants

- Attempt to complete representative task scenarios in an efficient manner.
- Complete post-task and post-test usability instruments honestly.

Ethics

All persons involved with the usability test are required to adhere to the following ethical guidelines:

- The performance of any test participant must not be individually attributable. Individual participant names will not appear outside the testing session.
- A description of the participant's performance will not be reported to their instructor or academic department head.
- All survey entries, task metrics, and user comments remain confidential and serve academic interface evaluation exclusively.

Usability Tasks

The usability tasks derive from student use cases and workflows identified during the heuristic evaluation of the MyMapúa portal. Because participants have limited evaluation time, the tasks focus on high-frequency academic activities: course schedule checking, grade breakdown verification, cumulative GPA inspection, degree curriculum tracking, and tuition billing review. All participants complete identical tasks.

The evaluation uses a responsive web prototype deployed on the Vercel hosting platform. The prototype operates on client-side state and mock student data, eliminating connections to production university databases. Testing activities produce zero impact on official student academic records or financial transactions.

The task descriptions below are required to be reviewed by the application owner, business-process owner, development owner, and/or deployment manager to ensure that the content, format, and presentation are representative of real use and substantially evaluate the total application. Their acceptance is to be documented prior to usability test.

The scenarios encompass the primary academic tracking workflows supported by the mobile portal: checking enrolled schedules, inspecting granular grade components, checking cumulative GPA or GWA, reviewing remaining graduation credits in the curriculum, and reviewing tuition charges on the Statement of Account.

- Task 1: Log in to the mobile prototype, navigate to your current term dashboard, and identify the course title, room assignment, and meeting time for your first scheduled class.
- Task 2: Navigate to the My Grades section, select your active software course, and locate the itemized score breakdown for laboratory submissions and quizzes.
- Task 3: Locate your cumulative Grade Point Average (GPA) or general weighted average (GWA) from the previous term on the mobile dashboard or academic record screen.
- Task 4: Open the My Curriculum view and determine the total number of credit units remaining for graduation.
- Task 5: From the dashboard or navigation menu, access your Statement of Account (SOA) and identify your outstanding balance and next payment due date.

Usability Metrics

Usability metrics refers to user performance measured against specific performance goals necessary to satisfy usability requirements. Scenario completion success rates, adherence to dialog scripts, error rates, and subjective evaluations will be used. Time-to-completion of scenarios will also be collected.

Scenario Completion

Each scenario will require, or request, that the participant obtains or inputs specific data that would be used in course of a typical task. The scenario is completed when the participant indicates the scenario's goal has been obtained (whether successfully or unsuccessfully) or the participant requests and receives sufficient guidance as to warrant scoring the scenario as a critical error.

Critical Errors

Critical errors are deviations at completion from the targets of the scenario. Obtaining or otherwise reporting of the wrong data value due to participant workflow is a critical error. Participants may or may not be aware that the task goal is incorrect or incomplete.

Independent completion of the scenario is a universal goal; help obtained from the other usability test roles is cause to score the scenario a critical error. Critical errors can also be assigned when the participant initiates (or attempts to initiate) an action that will result in the goal state becoming unobtainable. In general, critical errors are unresolved errors during the process of completing the task or errors that produce an incorrect outcome.

Non-critical Errors

Non-critical errors are errors that are recovered from by the participant or, if not detected, do not result in processing problems or unexpected results. Although non-critical errors can be undetected by the participant, when they are detected they are generally frustrating to the participant.

These errors may be procedural, in which the participant does not complete a scenario in the most optimal means (e.g., excessive steps and keystrokes). These errors may also be errors of confusion (ex., initially selecting the wrong function, using a user-interface control incorrectly such as attempting to edit an un-editable field).

Noncritical errors can always be recovered from during the process of completing the scenario. Exploratory behavior, such as opening the wrong menu while searching for a function, will be coded as a non-critical error.

Subjective Evaluations

Subjective evaluations regarding ease of use and satisfaction will be collected via questionnaires, and during debriefing at the conclusion of the session. The questionnaires will utilize free-form responses and rating scales.

Scenario Completion Time (time on task)

The time to complete each scenario, not including subjective evaluation durations, will be recorded.

Usability Goals

The next section describes the usability goals for the redesigned MyMapúa prototype.

Completion Rate

Completion rate is the percentage of test participants who successfully complete the task without critical errors. A critical error is defined as an error that results in an incorrect or incomplete outcome. In other words, the completion rate represents the percentage of participants who, when they are finished with the specified task, have an "output" that is correct. Note: If a participant requires assistance in order to achieve a correct output then the task will be scored as a critical error and the overall completion rate for the task will be affected.

A completion rate of 100% is the goal for each task in this usability test.

Error-free rate

Error-free rate is the percentage of test participants who complete the task without any errors (critical or non-critical errors). A non-critical error is an error that would not have an impact on the final output of the task but would result in the task being completed less efficiently.

An error-free rate of 80% is the goal for each task in this usability test.

Time on Task (TOT)

The time to complete a scenario is referred to as "time on task". It is measured from the time the person begins the scenario to the time he/she signals completion.

Subjective Measures

Subjective opinions about specific tasks, time to perform each task, features, and functionality will be surveyed. At the end of the test, participants will rate their satisfaction with the overall system. Combined with the interview/debriefing session, these data are used to assess attitudes of the participants.

Problem Severity

To prioritize recommendations, a method of problem severity classification will be used in the analysis of the data collected during evaluation activities. The approach treats problem severity as a combination of two factors - the impact of the problem and the frequency of users experiencing the problem during the evaluation.

Impact

Impact is the ranking of the consequences of the problem by defining the level of impact that the problem has on successful task completion. There are three levels of impact:

- High - prevents the user from completing the task (critical error)
- Moderate - causes user difficulty but the task can be completed (non-critical error)
- Low - minor problems that do not significantly affect the task completion (non-critical error)

Frequency

Frequency is the percentage of participants who experience the problem when working on a task.

- High: 30% or more of the participants experience the problem
- Moderate: 11% - 29% of participants experience the problem
- Low: 10% or fewer of the participants experience the problem

For a sample of 15 participants, experiencing an issue once corresponds to 6.7% frequency (Low), two to four occurrences correspond to 13.3% to 26.7% frequency (Moderate), and five or more occurrences represent 33.3% or higher frequency (High).

Problem Severity Classification

The identified severity for each problem implies a general reward for resolving it, and a general risk for not addressing it, in the current release.

Severity 1 - High impact problems that often prevent a user from correctly completing a task. They occur in varying frequency and are characteristic of calls to the Help Desk. Reward for resolution is typically exhibited in fewer Help Desk calls and reduced redevelopment costs.

Severity 2 - Moderate to high frequency problems with moderate to low impact are typical of erroneous actions that the participant recognizes needs to be undone. Reward for resolution is typically exhibited in reduced time on task and decreased training costs.

Severity 3 - Either moderate problems with low frequency or low problems with moderate frequency; these are minor annoyance problems faced by a number of participants. Reward for resolution is typically exhibited in reduced time on task and increased data integrity.

Severity 4 - Low impact problems faced by few participants; there is low risk to not resolving these problems. Reward for resolution is typically exhibited in increased user satisfaction.

Reporting Results

The Usability Test Report will be provided at the conclusion of the usability test. It will consist of a report and/or a presentation of the results; evaluate the usability metrics against the pre-approved goals, subjective evaluations, and specific usability problems and recommendations for resolution. The recommendations will be categorically sized by development to aid in implementation strategy. The report is anticipated to be delivered to the Project UCD Contact by October 2, 2026.

USABILITY SCRIPT AND SCORES

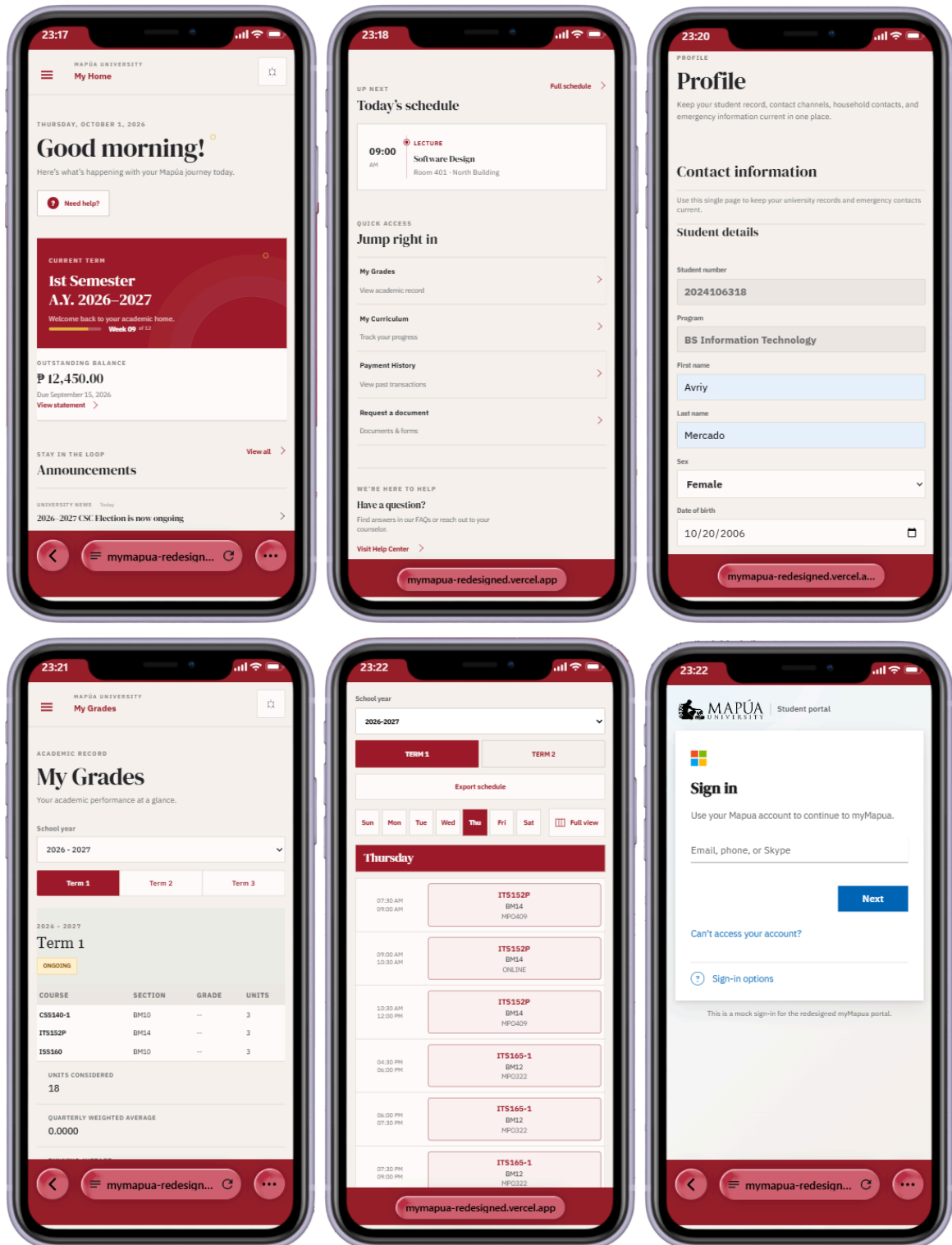
This section presents the usability testing script and scoring criteria used to evaluate the redesigned MyMapúa prototype. It includes the tasks given to participants and the scoring method used to measure task completion, errors, and overall usability.

Features & functionality			Weighting	Weighting ratio	Rating (0 - 5)	Score	Out of
1	The features and functions of the redesigned MyMapúa meet my needs as a Mapúa student.	Excellent	5	100%	4.60	4.60	5.00
2	The features and functions support the way I normally accomplish tasks in MyMapúa.	Excellent	5	100%	4.67	4.67	5.00
3	Frequently used tasks are readily available and easily accessible from the homepage.	Excellent	4	80%	4.67	3.73	4.00
4	The redesigned MyMapúa provides appropriate help and instructions for users when needed.	Excellent	3	60%	4.67	2.80	3.00
5	Buttons and call-to-action options are clearly labeled and appear clickable.	Excellent	3	60%	4.60	2.76	3.00
Homepage / starting page							
6	The homepage provides a clear overview of the content, features, and functionality available.	Excellent	3	60%	4.67	2.80	3.00
7	The homepage makes it easy for me to find the information or task I need.	Excellent	4	80%	4.67	3.73	4.00
8	The homepage layout is clear, organized, and has sufficient spacing between elements.	Excellent	3	60%	4.60	2.76	3.00
Navigation and Structure							
9	I can easily access the redesigned MyMapúa on my mobile device when needed.	Excellent	2	40%	4.53	1.81	2.00
10	The navigation menu is easy to find, intuitive, and consistent.	Excellent	4	80%	4.60	3.68	4.00
11	The navigation provides convenient ways for me to access the information and features I need.	Excellent	3	60%	4.53	2.72	3.00
12	The site or application structure is clear, easily understood and addresses common user goals.	Excellent	5	100%	4.67	4.67	5.00
13	Links and navigation options are clear, descriptive, and well labeled.	Excellent	3	60%	4.67	2.80	3.00
14	The mobile back or previous-page function works as expected.	Excellent	4	80%	4.73	3.79	4.00
15	My current location within the application is clearly indicated.	Excellent	2	40%	4.73	1.89	2.00
16	I can easily return to the homepage or another relevant starting point.	Excellent	2	40%	4.67	1.87	2.00
17	The available menu provides a clear and organized way to access the application's different sections.	Excellent	1	20%	4.67	0.93	1.00
Responsive and Adaptive Mobile Design							
18	The redesigned MyMapúa layout adjusts appropriately to my mobile screen size.	Excellent	4	80%	4.60	3.68	4.00
19	The content and interface elements are properly arranged and fit within my mobile screen.	Excellent	4	80%	4.80	3.84	4.00
20	The interface remains easy to use across different mobile screen sizes.	Excellent	2	40%	4.80	1.92	2.00
21	Text, buttons, icons, and other elements remain clear and accessible on my mobile device.	Excellent	4	80%	4.80	3.84	4.00
Feedback and User Control							
22	Prompt and appropriate feedback is provided after I complete an action.	Excellent	4	80%	4.73	3.79	4.00
23	I can easily go back, change, or cancel an action when necessary.	Excellent	3	60%	4.67	2.80	3.00
24	I can easily provide feedback or contact support when needed.	Excellent	1	20%	4.67	0.93	1.00
Forms and Processes							
25	Complex forms and processes are broken down into clear and understandable steps or sections.	Excellent	3	60%	4.67	2.80	3.00
26	Only the necessary information is requested when I complete a form or task.	Excellent	2	40%	4.67	1.87	2.00

27	Required and optional form fields are clearly indicated.	Excellent		2	40%	4.60	1.84	2.00
28	Appropriate input fields and selection options are provided for the information being requested.	Excellent		3	60%	4.73	2.84	3.00
29	Help and instructions are provided when necessary to complete a form or task.	Excellent		3	60%	4.67	2.80	3.00
Errors and Recovery								
30	Errors are clear, easily identifiable, and displayed in an appropriate location.	Good		4	80%	4.47	3.57	4.00
31	Error messages are concise, easy to understand, and explain what action I need to take.	Excellent		3	60%	4.53	2.72	3.00
32	Common user errors, such as missing or invalid information, are considered and prevented where	Excellent		3	60%	4.53	2.72	3.00
33	I can easily recover from errors without having to start the entire task again.	Excellent		3	60%	4.60	2.76	3.00
Content & text								
34	The content available is appropriate, relevant, and sufficiently detailed to meet my needs.	Excellent		5	100%	4.60	4.60	5.00
35	Useful and relevant links or related information are available and shown in context.	Excellent		2	40%	4.67	1.87	2.00
36	The language, terminology, and tone used are appropriate and easy for me to understand.	Excellent		4	80%	4.73	3.79	4.00
37	The terms, language, and tone used throughout the application are consistent.	Excellent		3	60%	4.60	2.76	3.00
38	The text and content are legible and easy to scan, with good typography and visual contrast.	Excellent		3	60%	4.73	2.84	3.00
Help and Support								
39	Help and instructions are provided in language that is easy for me to understand.	Excellent		4	80%	4.80	3.84	4.00
40	The available help is concise, easy to read, and easy to understand.	Excellent		3	60%	4.67	2.80	3.00
41	Accessing help does not prevent me from returning to or continuing my current task.	Excellent		3	60%	4.67	2.80	3.00
42	I can easily get further help or support when needed.	Excellent		2	40%	4.73	1.89	2.00
Performance								
43	The redesigned MyMapúa performs efficiently without slow loading or unnecessary delays.	Excellent		4	80%	4.73	3.79	4.00
44	Errors and reliability issues do not interfere with my experience when using the redesigned MyMapúa.	Excellent		4	80%	4.67	3.73	4.00
45	The redesigned MyMapúa works reliably while I am completing my tasks.	Excellent		3	60%	4.67	2.80	3.00
Overall weighted usability score (out of 100) *		93.2	– Excellent	5		134.2	144	

* Very poor (less than 29) - Users are likely to experience very significant difficulties using this site or system and might
 * Poor (between 29 and 49) - Users are likely to experience some difficulties using this site or system and might not be
 * Moderate (between 49 and 69) - Users should be able to use this site or system and complete most important tasks,
 * Good (between 69 and 89) - Users should be able to use this site or system with relative ease and should be able to
 * Excellent (more than 89) - This site or system provides an excellent user experience for users. Users should be able to

Screenshots:



USABILITY REPORT

The table shows the average scores for each interface category of the redesigned MyMapúa prototype, based on 15 testers' ratings (0–5). It also presents the category averages, weighted scores, percentages, and corresponding interpretations.

Category	Items	Avg. Rating (0-5)	Weighted Score	Score %	Interpretation
Features & Functionality	1–5	4.64	18.57 / 20	92.8%	Excellent
Homepage / Starting Page	6–8	4.65	9.30 / 10	93.0%	Excellent
Navigation & Structure	9–17	4.64	24.15 / 26	92.9%	Excellent
Responsive & Adaptive Mobile Design	18–21	4.75	13.28 / 14	94.9%	Excellent
Feedback & User Control	22–24	4.69	7.51 / 8	93.9%	Excellent
Forms & Processes	25–29	4.67	12.15 / 13	93.5%	Excellent
Errors & Recovery	30–33	4.53	11.78 / 13	90.6%	Excellent
Content & Text	34–38	4.67	15.85 / 17	93.2%	Excellent
Help & Support	39–42	4.72	11.33 / 12	94.4%	Excellent
Performance	43–45	4.69	10.32 / 11	93.8%	Excellent
Overall	1–45	4.66	134.2 / 144	93.2%	Excellent

Table 1. Average scores and interpretation per category

Rating range (score %)	Interpretation
Very Poor (less than 29)	Users are likely to experience very significant difficulties using the site or system.
Poor (29 to 49)	Users are likely to experience some difficulties and might not be able to complete some tasks.
Moderate (49 to 69)	Users should be able to use the system and complete most important tasks.
Good (69 to 89)	Users should be able to use the system with relative ease.
Excellent (more than 89)	The system provides an excellent user experience for users.

Table 2. Rating ranges used for interpretation

Interpretation

Features & Functionality (Avg. 4.64; 92.8%; Excellent)

Participants agreed that the prototype's features meet their needs as students and support how they normally work (items 1 and 2 averaged 4.60 and 4.67). Frequently used tasks were rated as easy to reach from the homepage (4.67), and buttons were seen as clearly labeled and clickable (4.60). The category is Excellent, and the slightly lower item 1 and item 5 ratings suggest room to confirm that no needed feature or label is missing.

Homepage / Starting Page (Avg. 4.65; 93.0%; Excellent)

The homepage was rated as giving a clear overview of available content and as making it easy to find needed information (both 4.67). Layout clarity and spacing scored 4.60. This supports the redesign decision to surface the schedule, balance, and quick-access links on the home screen.

Navigation & Structure (Avg. 4.64; 92.9%; Excellent)

Navigation was rated Excellent overall, with the back function and the indication of current location the highest items (both 4.73). The lowest items were easy access on a mobile device (4.53), convenient access to needed information (4.53), and an intuitive, consistent menu (4.60). Navigation carries the largest number of items (9) and many of the highest weights, so small gains here have a noticeable effect on the overall score.

Responsive & Adaptive Mobile Design (Avg. 4.75; 94.9%; Excellent)

This was the highest-scoring category. Layout fit, consistency across screen sizes, and legibility of text, buttons, and icons each averaged 4.80, while adjustment to the user's screen size scored 4.60. These results directly support the project's mobile-first objective and indicate the layout, spacing, and touch-target changes were effective.

Feedback & User Control (Avg. 4.69; 93.9%; Excellent)

Participants found that the prototype gives prompt, appropriate feedback after actions (4.73) and lets them go back, change, or cancel actions (4.67). The ability to give feedback or contact support was also rated 4.67. The category is Excellent.

Forms & Processes (Avg. 4.67; 93.5%; Excellent)

Forms and processes were rated consistently high, with appropriate input fields and selection options the strongest item (4.73). Clear division of processes into steps, requesting only necessary information, and help for completing tasks each scored 4.67, and indication of required versus optional fields scored 4.60.

Errors & Recovery (Avg. 4.53; 90.6%; Excellent)

This was the lowest-scoring category, although it remains in the Excellent range. Item 30, on errors being clear, identifiable, and well placed, averaged 4.47 and is the only item in the instrument that falls below the 4.50 level. Error message clarity (item 31) and error prevention (item 32) both averaged 4.53, and ease of recovery (item 33) averaged 4.60. Because the prototype runs on mock data, participants likely encountered few real errors, so these ratings may reflect first impressions of error handling rather than extensive experience.

Content & Text (Avg. 4.67; 93.2%; Excellent)

Content was rated as relevant and sufficiently detailed (4.60), consistent in terminology and tone (4.60), appropriate and easy to understand (4.73), and legible with good contrast (4.73). Useful links and related information in context averaged 4.67.

Help & Support (Avg. 4.72; 94.4%; Excellent)

Help written in easy-to-understand language received the highest rating in the category (4.80). Concise help content, help that does not interrupt the current task (both 4.67), and ease of getting further support (4.73) were also rated Excellent.

Performance (Avg. 4.69; 93.8%; Excellent)

Participants judged the prototype to load efficiently (4.73) and to work reliably during tasks (4.67), and reported that errors or reliability issues did not interfere with their experience (4.67). Note that these ratings apply to a lightweight, client-side prototype and should be re-tested against production data and network load.

Overall Interpretation

The overall weighted usability score is 134.2 out of 144 (93.2%), with a mean rating of 4.66 out of 5. This is in the Excellent range (more than 89), as are all ten categories. Responsive & Adaptive Mobile Design scored highest (94.9%) and Errors & Recovery scored lowest (90.6%), so error messaging is the main area to refine.

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APPENDICES

Appendix A. Redesigned MyMapúa Mobile Web Link

<https://mymapua-redesigned.vercel.app/>



Appendix B. User Usability Questionnaire

MyMapua Mobile Redesigned Usability Test

Good day! We are a group of IT students from Mapúa University conducting an academic study to evaluate our redesigned MyMapúa mobile interface.

This usability test is focused specifically on evaluating the **usability and user experience** of the redesigned MyMapúa mobile interface. Participants will be asked to explore the completed prototype and perform selected tasks based on common MyMapúa functions.

Why your feedback matters:
Your responses will help us determine how easy and convenient the redesigned interface is to use. The evaluation will focus on factors such as **ease of use, navigation, clarity, efficiency, functionality, and overall user satisfaction.**

Important Note:
This is only a **usability test** for the redesigned MyMapúa mobile interface. You **do not need to create your own account or provide your personal MyMapúa login credentials.** All necessary access or test information will be provided by the researchers.

Data Privacy Notice:
This test will take approximately **5-10 minutes** to complete. All information gathered will be kept confidential and will be used solely for **academic and research purposes.** Your responses will only be used to evaluate the usability of the redesigned interface.

Website/Prototype Link:
<https://mymapua-redesigned.vercel.app/>

Use these credentials: Email: student@gmail.com Password: password123

Researchers/Developers:
De Dios, Robert Cesar Mendocsa, Miguel Mercado, Arny Disneffomero, Ibrahim Ryan Acor II

Institution: Mapúa University

Program: Bachelor of Science in Information Technology

Thank you for your time and contribution to our study!

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

1. Have you already viewed and tested the redesigned MyMapúa mobile interface? *

☐ Yes

☐ No

Next

Microsoft 365

This content is created by the owner of the form. The data you submit will be sent to the form owner. Microsoft is not responsible for the privacy or security practices of its customers, including those of this form owner. Never give out your password.

MyMapua Mobile Redesigned Usability Test

* Required

Usability Evaluation

Please rate the following statements based on your experience using the redesigned MyMapúa interface.

Note: This test is for **usability evaluation only.** Please **do not create a new account or use your personal MyMapúa credentials.** Use the provided email and password to access the system and complete the assigned tasks.

3. **Features & Functionality** *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
The features and functions of the redesigned MyMapúa meet my needs as a Mapúa student.	☐	☐	☐	☐	☐
The features and functions support the way I normally accomplish tasks on MyMapúa.	☐	☐	☐	☐	☐
Frequently used tasks are readily available and easily accessible from the homepage.	☐	☐	☐	☐	☐
The redesigned MyMapúa provides appropriate help and instructions for users when needed.	☐	☐	☐	☐	☐
Buttons and call-to-action options are clearly labeled and appear clickable.	☐	☐	☐	☐	☐

Back **Next**

Microsoft 365

This content is created by the owner of the form. The data you submit will be sent to the form owner. Microsoft is not responsible for the privacy or security practices of its customers, including those of this form owner. Never give out your password.

Microsoft Forms | AI-Powered surveys, quizzes and polls. [Create your own today.](https://forms.office.com)

The owner of this form has not provided a privacy statement as to how they will use your response data. Do not provide personal or sensitive information. [Terms of use](#)

MyMapua Mobile Redesign Usability Test

* Required

Usability Evaluation

Please view the following statements based on your experience using the redesigned MyMapua interface.

Note: This test is for **usability evaluation only**. These do not create a new account or use your personal MyMapua credentials. Use the provided email and password to access the system and complete the assigned tasks.

7. Feedback and User Control

	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
Diagrams and appropriate feedback is provided after completing an action.	☐	☐	☐	☐	☐
I can easily go back, change or cancel an action when necessary.	☐	☐	☐	☐	☐
I am given an appropriate confirmation notice upon completing an important action.	☐	☐	☐	☐	☐
I can easily provide feedback or cancel support when needed.	☐	☐	☐	☐	☐

8. Forms and Processes

	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
Complex forms and processes are broken down into clear and understandable steps or actions.	☐	☐	☐	☐	☐
Only the necessary information is requested when completing a form or task.	☐	☐	☐	☐	☐
Required and optional form fields are clearly indicated.	☐	☐	☐	☐	☐
Appropriate focus fields and selection options are provided for the information being requested.	☐	☐	☐	☐	☐
Help and instructions are provided when necessary to complete a form or task.	☐	☐	☐	☐	☐

MyMapua Mobile Redesigned Usability Test

*** Required**

Usability Evaluation

Please rate the following statements based on your experience using the redesigned MyMapua interface.

Notes: This test is for **usability evaluation only**. Please do not create a new account or use your personal MyMapua credentials. Use the provided email and password to access the system and complete the assigned tasks.

11. Help and Support *

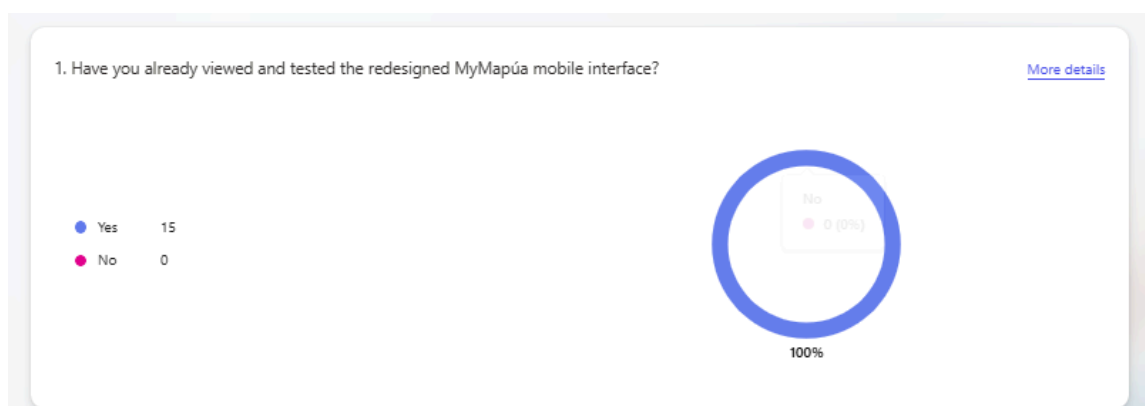
	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
Help and instructions are provided in language that is easy for me to understand.	☐	☐	☐	☐	☐
The available help is concise, easy to read, and easy to understand.	☐	☐	☐	☐	☐
Accessing help does not prevent me from returning to or continuing my current task.	☐	☐	☐	☐	☐
I can easily get further help or support when needed.	☐	☐	☐	☐	☐

12. Performance and Reliability *

	Strongly Disagree	Disagree	Neutral	Agree	Strongly agree
The redesigned MyMapua performs efficiently without slow loading or unnecessary delays.	☐	☐	☐	☐	☐
Errors and reliability issues do not interfere with my experience when using the redesigned MyMapua.	☐	☐	☐	☐	☐
The redesigned MyMapua works reliably while I am completing my tasks.	☐	☐	☐	☐	☐

Back Continue

Appendix C. User Usability Survey Results



3. Features & Functionality

[More details](#)

● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

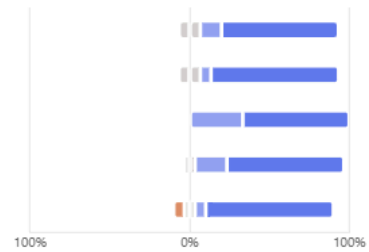
The features and functions of the redesigned MyMapúa meet my needs as a Mapúa student.

The features and functions support the way I normally accomplish tasks in MyMapúa.

Frequently used tasks are readily available and easily accessible from the homepage.

The redesigned MyMapúa provides appropriate help and instructions for users when needed.

Buttons and call-to-action options are clearly labeled and appear clickable.



4. Homepage / Starting Page

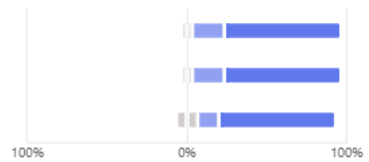
[More details](#)

● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

The homepage provides a clear overview of the content, features, and functionality available.

The homepage makes it easy for me to find the information or task I need.

The homepage layout is clear, organized, and has sufficient spacing between elements.



5. Navigation and Structure

[More details](#)

● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

I can easily access the redesigned MyMapúa on my mobile device when needed.

The navigation menu is easy to find, intuitive, and consistent.

The navigation provides convenient ways for me to access the information and features I need.

The structure of the redesigned MyMapúa is clear, easy to understand, and supports my common tasks.

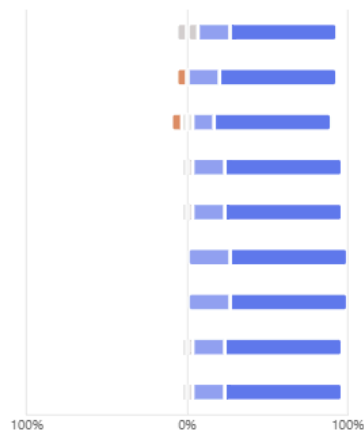
Links and navigation options are clear, descriptive, and well labeled.

The mobile back or previous-page function works as expected.

My current location within the application is clearly indicated.

I can easily return to the homepage or another relevant starting point.

The available menu provides a clear and organized way to access the application's different sections.



6. Responsive and Adaptive Mobile Design

[More details](#)

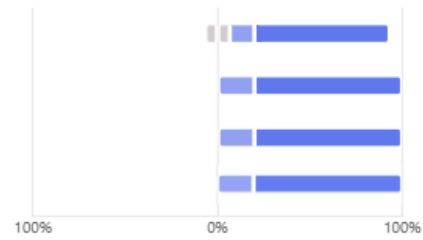
● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly Agree

The redesigned MyMapúa layout adjusts appropriately to my mobile screen size.

The content and interface elements are properly arranged and fit within my mobile screen.

Text, buttons, icons, and other elements remain clear and accessible on my mobile device.

The interface remains easy to use across different mobile screen sizes.



7. Feedback and User Control

[More details](#)

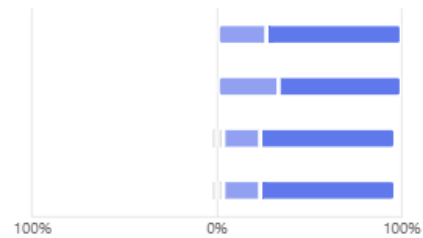
● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

Prompt and appropriate feedback is provided after I complete an action.

I can easily go back, change, or cancel an action when necessary.

I am given an appropriate confirmation before completing an important action.

I can easily provide feedback or contact support when needed.



8. Forms and Processes

[More details](#)

● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

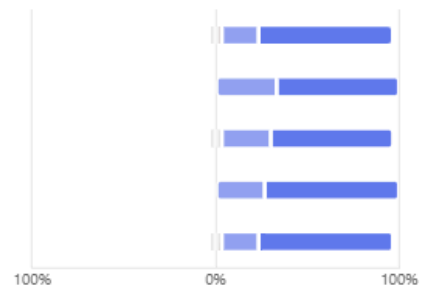
Complex forms and processes are broken down into clear and understandable steps or sections.

Only the necessary information is requested when I complete a form or task.

Required and optional form fields are clearly indicated.

Appropriate input fields and selection options are provided for the information being requested.

Help and instructions are provided when necessary to complete a form or task.



9. Errors and Recovery

[More details](#)

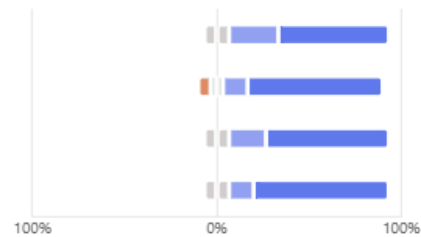
● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly Agree

Errors are clear, easily identifiable, and displayed in an appropriate location.

Error messages are concise, easy to understand, and explain what action I need to take.

Common user errors, such as missing or invalid information, are considered and prevented where possible.

I can easily recover from errors without having to start the entire task again.



10. Content

[More details](#)

● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

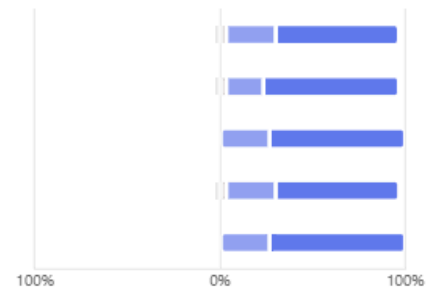
The content available is appropriate, relevant, and sufficiently detailed to meet my needs.

Useful and relevant links or related information are available and shown in context.

The language, terminology, and tone used are appropriate and easy for me to understand.

The terms, language, and tone used throughout the application are consistent.

The text and content are legible and easy to scan, with good typography and visual contrast.



11. Help and Support

[More details](#)

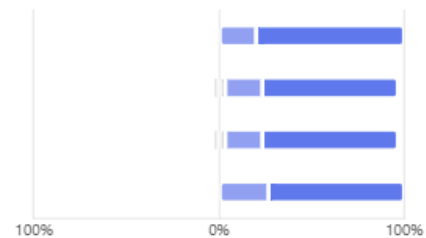
● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

Help and instructions are provided in language that is easy for me to understand.

The available help is concise, easy to read, and easy to understand.

Accessing help does not prevent me from returning to or continuing my current task.

I can easily get further help or support when needed.



12. Performance and Reliability

[More details](#)

● Strongly Disagree ● Disagree ● Neutral ● Agree ● Strongly agree

The redesigned MyMapúa performs efficiently without slow loading or unnecessary delays.

Errors and reliability issues do not interfere with my experience when using the redesigned MyMapúa.

The redesigned MyMapúa works reliably while I am completing my tasks.

